Holdout

Offers only where they change the outcome.

Bot @navpreet_offers_bot
Dashboard offer-engine-nu.vercel.app
Repo github.com/Navkaur-codes/Offer-Engine

Online shops send discounts to everyone and count every sale that follows as a win, but many of those customers would have bought anyway, so the discount simply gives away margin. **Holdout decides, for each customer and each moment, whether an offer will actually change what they do, and only then which offer and when.** Offers go out through a live Telegram bot; marketers set the limits and see the proof on a web dashboard. The name comes from the holdout: the group that deliberately gets nothing, which is how we measure what offers really caused.

+₹91,760	₹14,916	−₹24,320	100%
incremental revenue vs the holdout, 12 months	margin saved by not discounting sure-things	lost by blanket free shipping: why targeting matters	of offers re-checked by one rule validator

How one decision is made

Event	Pre-gate	Policy	AI agent	Validator	Sent + logged

- **Pre-gate**, plain code before any AI: kill switch, `/stop` and `/pause`, weekly frequency, budget, no discount within 48 h of a purchase, quiet hours (21:00–09:00 for proactive messages), and the 10% holdout.
- **Policy**: the customer's live profile (churn risk, uplift, behaviour) places them in one of 10 policies; the policy decides which offers and caps are allowed (page 2). Layers Global → Festival → Persona → Individual can only tighten.
- **AI agent** (Groq gpt-oss-120b, read-only tools, strict JSON): picks the best allowed offer and writes a message of at most 280 characters. Customer-typed text is treated as data, never as instructions.
- **Validator**: the **same** `evaluate()` **function** re-checks the exact offer and every number in the message. If it fails twice, or the AI is slow, down or over budget, a **rules fallback** decides, so the bot is never silent and never breaks a rule.
- **Logged with a reason**: every decision, including "no offer", is stored with why this customer, why this offer, the alternatives rejected and the rules applied. Explanations come from rule results, never the AI's words.

The machine learning behind it

Model	What it does	How
Churn risk	probability of no purchase in the next 60 days	logistic regression, 7 behaviour features, AUC ≈ 0.78
Uplift (T-learner)	how much an offer changes the chance to buy	one model on customers who got past offers, one on the 10% who didn't; the difference is the uplift (gradient-boosted trees, Qini-checked)
Expected incremental profit	is the offer worth it?	uplift × basket × category margin − cost, where the cost counts the discount for **everyone who buys**, including people who'd buy anyway. Free perks first.
Experiments	did it work?	each campaign compared with its own 10% control group: revenue, profit, ROI by offer type, Qini

What you can see and use

Part	Highlights
Telegram bot	`/start` (3 taps build a persona), `/simulate cart · browse · purchase · inactive · festival · payday`, offers as **picture cards** with ✓ Claim · ⏰ Remind me · 🙈 Not for me · ? Why this?, `/why /history /pause /stop`
Customer Lab	a Telegram group where 10 synthetic personas, voiced by "Lab Shopper" bots, shop and react; judges watch different customers get different offers, and some get nothing on purpose
Dashboard	Overview (live stream, hero KPI), Decisions with the Pipeline Lens, Rules as editable sentences with impact preview and revert, Customers, Experiments, Festivals, Realism, Playground with "force LLM failure"
Safety & culture	kill switch, consent at once, quiet hours; festival greetings only for customers who opted in (religion never inferred from a name); restricted promotions blocked

Built with Python · FastAPI · scikit-learn · PostgreSQL · Groq · python-telegram-bot · Pillow · React + TypeScript · hosted on Render, Neon and Vercel · 362 automated tests.

Every synthetic customer has **two layers**: a **hidden response type** (how an offer would really affect them; the engine never sees it and must infer it from behaviour, as in real life) and a **visible persona** that shapes their shopping. From live behaviour, the engine assigns one of **10 policies**, which sets the offers they may receive.

1 · Hidden response types: the four kinds of customer

Type	Share	With an offer	Without	What Holdout does
Persuadable	~40%	buys	doesn't	**makes an offer**, the only group where an offer creates new revenue
Sure-thing	~22%	buys	buys	**doesn't pay them**: a discount would only give away margin (₹14,916 saved)
Lost cause	~27%	doesn't	doesn't	doesn't waste budget; at most a free perk
Sleeping dog	~11%	**doesn't**	buys	**never contacts them**: offers put them off (needs negative uplift and a past "Not for me")

2 · Synthetic personas (how they shop)

Persona	Share	Typical behaviour	Hidden-type mix
Loyal VIP	6%	frequent, high value, full price	70% sure-thing · 15% persuadable · 10% sleeping dog
Deal hunter	16%	buys with coupons, compares prices	50% persuadable · 30% sure-thing
At-risk regular	10%	was regular, now buying less	55% persuadable · 25% lost cause
New user	10%	recent sign-up, 0–1 orders	45% persuadable · 25% lost cause
Dormant	12%	no activity for months	55% lost cause · 25% persuadable
Window shopper	16%	many views, rarely buys	45% lost cause · 30% persuadable
Cart abandoner	12%	adds to cart, leaves	60% persuadable · 20% lost cause
Festive gifter	12%	spikes around festivals, gifting	40% sure-thing · 35% persuadable
High returner	6%	returns many orders	30% sure-thing · 25% sleeping dog

3 · The 10 live policies (assigned from behaviour, in this order)

Policy	Assigned when	Offers allowed / limits
High returner	return rate ≥ 40% over ≥ 3 orders	points only (bonus points, points reminders)
Sleeping dog	negative uplift for every offer and ≥ 1 "Not for me"	**no proactive offers at all**
VIP	top 10% lifetime value, bought in the last 30 days, ≥ 3 orders	early access, bonus points, free shipping or express; **never a discount**
Cart abandoner	open cart 30 min – 72 h old	ladder: cart reminder or free shipping, then up to 5% off after 24 h
Dormant	no activity for 120+ days	one try every 90 days; up to 15% off, free shipping, reminder
At-risk	churn ≥ 0.6 and 30+ days quiet, or 60+ days quiet; ≥ 2 orders and ₹2,500+ spent	win-back only if the offer lifts buying by > 5 points; up to 20% off; once in 30 days
New user	joined ≤ 30 days ago, ≤ 1 order	up to 15% off, free shipping, reminders
Deal hunter	≥ 60% of orders used a coupon, ≥ 2 orders	at most 10% off and only on baskets above ₹999; flat ₹ off
Window shopper	≥ 8 views in 90 days, no orders	free shipping, early access, reminders, at most ₹150 off
Default	everyone else	global limits only

Every policy also sits inside the global limits: discount ≤ category margin − 5 points, at most 30% / ₹500 / 3× points, a daily budget, no % or ₹ discount within 48 h of a purchase, quiet hours, and messages ≤ 280 characters whose numbers must match the offer. Policies are recomputed on every event, so a customer who stops buying moves from VIP to at-risk and gets different rules.

4 · Customer Lab: ten personas live in Telegram

Arjun · VIP	Priya · Deal hunter	Gurpreet · At-risk regular	Simran · New shopper	Rohan · Cart abandoner
Aisha · Window shopper	Vikram · Frequent returner	Meera · **Sleeping dog**	Karan · Dormant	Neha · Festive gifter

Run `/lab run 3` in the group: Arjun gets early access (never a discount), Priya a capped discount over ₹999, Vikram points only, and Meera gets **nothing**: "no proactive offers for this customer".